
Significance Testing Algorithms

Simultaneous hypothesis testing was a lively topic in the early 1960s, my graduate student years, and had been so since the end of World War II. Rupert Miller's book *Simultaneous Statistical Inference* appeared in 1966, providing a beautifully lucid summary of the contemporary methodology. A second edition in 1981 recorded only modest gains during the intervening years. This was a respite, not an end: a new burst of innovation in the late 1980s generated important techniques that we will be revisiting in this chapter.

Miller's book, which gives a balanced picture of the theory of that time, has three notable features:

- 1 It is overwhelmingly frequentist.
- 2 It is focused on control of α , the overall Type I error rate of a procedure.¹
- 3 It is aimed at multiple testing situations with individual cases N between 2 and, say, 10.

We have now entered a scientific age in which $N = 10\,000$ is no cause for raised eyebrows. It is impressive (or worrisome) that the theory of the 1980s continues to play a central role in microarray-era statistical inference. Features 1 and 2 are still the norm in much of the multiple testing literature, despite the obsolescence of Feature 3. This chapter reviews part of that theory, particularly the ingenious algorithms that have been devised to control the overall Type I error rate (also known as FWER, the family-wise error rate). False discovery rate control, an approach which doesn't follow either Feature 1 or 2 and is better suited to the $N = 10\,000$ era, is taken up in Chapter 4. The material in this chapter is a digression from our chosen theme of empirical Bayes methods, and may be read lightly by those eager to get on with the main story.

¹ And, by extension, the construction of simultaneous confidence regions that have a guaranteed probability of containing *all* the relevant parameters.

3.1 *p*-Values and *z*-Values

First consider the classical “single-test” situation: we wish to test a single null hypothesis H_0 on the basis of observed data $\mathbf{x}$. For any value of α between 0 and 1 we construct a rejection region $\mathcal{R}_\alpha$ in the sample space of $\mathbf{x}$ such that

$$\Pr_0\{\mathbf{x} \in \mathcal{R}_\alpha\} = \alpha \quad [\alpha \in (0, 1)], \quad (3.1)$$

where $\Pr_0$ refers to the probability distribution of $\mathbf{x}$ under H_0 . The regions $\mathcal{R}_\alpha$ decrease with α ,

$$\mathcal{R}_\alpha \supseteq \mathcal{R}_{\alpha'} \quad \text{for } \alpha > \alpha'. \quad (3.2)$$

The *p*-value $p(\mathbf{x})$ corresponding to $\mathbf{x}$ is defined as the smallest value of α such that $\mathbf{x} \in \mathcal{R}_\alpha$,

$$p(\mathbf{x}) = \inf_{\alpha} \{\mathbf{x} \in \mathcal{R}_\alpha\}. \quad (3.3)$$

Intuitively, the smaller is $p(\mathbf{x})$, the more decisively is H_0 rejected. There is a more-or-less agreed-upon scale for interpreting *p*-values, originally due to Fisher, summarized in Table 3.1. Using *p*-values instead of a fixed rule like “reject at the $\alpha = 0.05$ level” is a more informative mode of data summary.

Table 3.1 *Fisher’s scale of evidence for interpreting p-values; for instance, $p(\mathbf{x}) = 0.035$ provides moderate to substantial grounds for rejecting H_0 .*

α	.10	.05	.025	.01	.001
Evidence against H_0 :	borderline	moderate	substantial	strong	overwhelming

For any value of u in $(0, 1)$, the event $p(\mathbf{x}) \leq u$ is equivalent to $\mathbf{x} \in \mathcal{R}_u$, implying

$$\Pr_0\{p(\mathbf{x}) \leq u\} = \Pr_0\{\mathbf{x} \in \mathcal{R}_u\} = u. \quad (3.4)$$

In other words, under H_0 the random variable $P = p(\mathbf{x})$ has a uniform distribution² over the interval $(0, 1)$,

$$H_0 : P = p(\mathbf{x}) \sim \mathcal{U}(0, 1). \quad (3.5)$$

² Here we are ignoring discrete null distributions, like binomial $(n, \frac{1}{2})$, where there are minor difficulties with definition (3.3).

P -values serve as a universal language for hypothesis testing. They allow for general rules of interpretation, such as Fisher's scale, applying to all hypothesis-testing situations. Fisher's famous $\alpha = 0.05$ dictum for "significance" has been overused and abused, but has served a crucial purpose nevertheless in bringing order to scientific reporting.

In subsequent chapters we will be working mainly with z -values rather than p -values,

$$z(\mathbf{x}) = \Phi^{-1}(p(\mathbf{x})), \quad (3.6)$$

where Φ^{-1} is the inverse function of the standard normal cdf, as in (2.5). Z -values also enjoy a universal null hypothesis distribution,

$$H_0 : z(\mathbf{x}) \sim \mathcal{N}(0, 1), \quad (3.7)$$

called the *theoretical null* at (2.6).

Figure 3.1 shows p -values and z -values for the DTI data, as partially reported in Figure 2.4. Here there are $N = 15\,443$ p -values p_i and likewise 15 443 z -values z_i , obtained from voxel-wise two-sample t -tests. The t -tests each have 10 degrees of freedom, so

$$p_i = F_{10}(t_i) \quad \text{and} \quad z_i = \Phi^{-1}(F_{10}(t_i)) \quad (3.8)$$

in the notation of (2.4)–(2.5). There are some interesting comparisons between the two displays:

- Both histograms show sharp discrepancies from their theoretical null distributions (3.5) or (3.7): the dramatic right spike of the p_i 's, and the corresponding heavy right tail of the z_i 's.
- The spike is more striking to the eye, but pays the price of collapsing all of the detail evident in the z -value tail.
- There are also *central* discrepancies between the theoretical null distributions and the histograms: this is clearer in the bottom panel, where the histogram center is shifted a little to the left of the $\mathcal{N}(0, 1)$ curve.

Exercise 3.1 How does the shift effect appear in the p -value histogram?

- The z_i 's, as monotone functions of the t_i 's in (3.8), automatically maintain the signs of the t -tests, with positive effects mapped to the right and negative effects to the left; definition (3.8) does the same for the p -values by mapping large positive effects toward $p = 1$ and negative effects toward $p = 0$. (Of course we would transform $p_i = 0.98$ to 0.02, etc., for interpretation in Fisher's scale.)

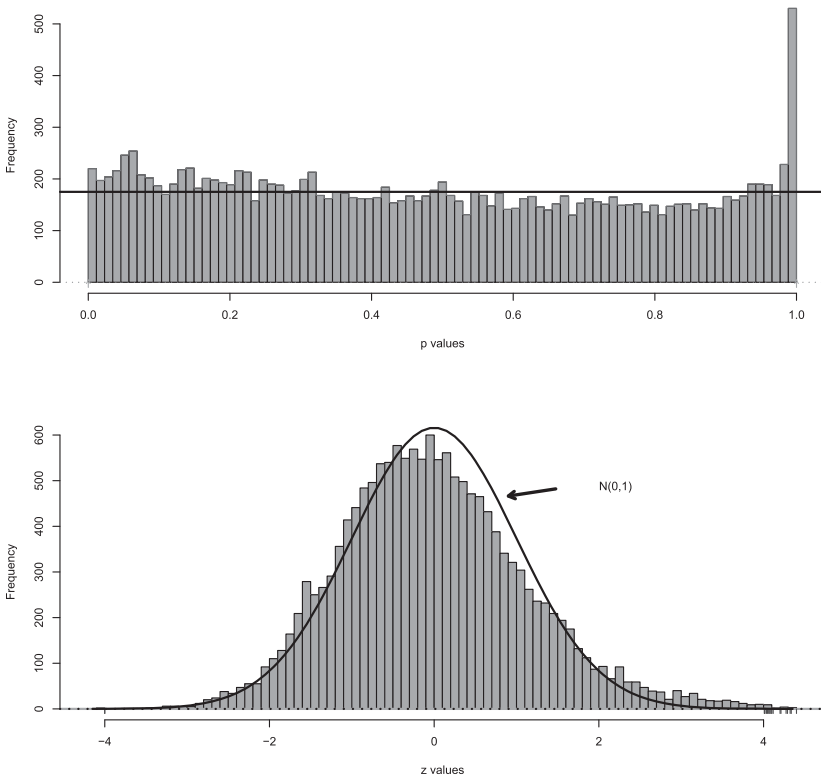


Figure 3.1 $N = 15\,443$ p -values (top) and z -values (bottom) for the DTI study partially displayed in Figure 2.4. Horizontal line in top panel indicates $\mathcal{U}(0, 1)$ distribution; curve in bottom panel indicates theoretical $\mathcal{N}(0, 1)$ null distribution. Small dashes at bottom right show the 18 z -values exceeding 4.

The literature tends to favor *two-sided* p -values, in our case

$$p_i^{(\text{two})} = \Pr \{ |T_{10}| > |t_i| \}, \quad (3.9)$$

where T_{10} indicates a standard Student- t variate with 10 degrees of freedom. Besides losing the sign information in t_i , this is a potentially dangerous definition if the test statistic, unlike T_{10} , has an asymmetric null distribution. A better option, if two-sided p -values are essential, is to transform our first definition to

$$p_i^{(\text{two})} = 2 \cdot \min(p_i, 1 - p_i), \quad (3.10)$$

agreeing with (3.9) in the symmetric but not the asymmetric case. Two-

sided testing makes some sense, sometimes, in classical single-test applications. It is less defensible in large-scale testing where, as in Figure 3.1, we can see the two sides behaving differently.

The author finds z -value histograms more informative than p -value histograms, but that is not the reason for the predominance of z -values in succeeding chapters: z -values allow us to bring the full power of normal theory to bear on large-scale inference problems. The `locfdr` algorithm of Chapter 5, for example, uses normal theory to estimate that the best-fit curve to the central peak in Figure 3.1 is $\mathcal{N}(-0.12, 1.06^2)$ rather than $\mathcal{N}(0, 1)$.

The original data matrix X for the DTI data is $15\,443 \times 12$, from which we hope to identify brain regions involved in dyslexia. In what follows we will be using a two-step analysis strategy: first we reduce each row of X to a single number p_i (or z_i); we then employ a testing algorithm to determine which of the N p_i values indicates non-null activity. Other methods that assume ANOVA-like structures for X are at least theoretically more informative, but I prefer the two-step approach. In my experience these huge data matrices show little respect for traditional analysis of variance structures. Using two-sample t -statistics t_i (or, if necessary, their Wilcoxon counterparts) as a starting point puts less strain on the statistical modeling. Nevertheless, modeling difficulties remain, as discussed in Chapter 10.

The use of p -values or z -values is not limited to two-sample situations. Suppose for example that y_j is some response variable, like a survival time, measured on subject j , $j = 1, 2, \dots, n$. For each row i of X we can run a linear regression of y_j on x_{ij} , calculate the slope coefficient $\hat{\beta}_i$, and take p_i to be its attained significance level, computed in the usual normal-theory way. (This requires modeling across individual rows of X , but not over the entire matrix.) Some of the examples to come involve more elaborate versions of this tactic.

3.2 Adjusted p -Values and the FWER

The family-wise error rate FWER is defined as the probability of making at least one false rejection in a family of hypothesis-testing problems. Let $p_1, p_2, \dots, p_N$ be the p -values obtained in tests of the corresponding family of null hypotheses $H_{01}, H_{02}, \dots, H_{0N}$. For the DTI data, N equals 15 443, with H_{0i} being the hypothesis of no response distribution difference between dyslexics and controls at voxel i , and p_i the observed p -value for the two-sample procedure testing for that difference. Some, perhaps most, of the null hypotheses H_{0i} will be true.

The family-wise error rate is

$$\text{FWER} = \Pr\{\text{Reject any true } H_{0i}\}; \quad (3.11)$$

a FWER control procedure is an algorithm that inputs a family of p -values $(p_1, p_2, \dots, p_N)$ and outputs the list of accepted and rejected null hypotheses, subject to the constraint

$$\text{FWER} \leq \alpha \quad (3.12)$$

for any preselected value of α .

Bonferroni's bound provides the classic FWER control method: we reject those null hypotheses for which

$$p_i \leq \alpha/N. \quad (3.13)$$

Let I_0 index the true null hypotheses, having N_0 members. Then

$$\begin{aligned} \text{FWER} &= \Pr\left\{\bigcup_{I_0}\left(p_i \leq \frac{\alpha}{N}\right)\right\} \leq \sum_{I_0} \Pr\left\{p_i \leq \frac{\alpha}{N}\right\} \\ &= N_0 \frac{\alpha}{N} \leq \alpha, \end{aligned} \quad (3.14)$$

verifying the FWER control property.³ The crucial step in the top line follows from Boole's inequality $\Pr\{\bigcup A_i\} \leq \sum \Pr\{A_i\}$.

One way to think about the Bonferroni bound is that the individual p -value p_i for testing H_{0i} translates into the family-wise *adjusted p -value*

$$\tilde{p}_i = \{Np_i\}_1 \quad (3.15)$$

(where $\{x\}_1$ is short for $\min(x, 1)$); we reject H_{0i} in the family-wise context if $\tilde{p}_i$, rather than p_i , is less than α . Though not of much use here, the language of adjusted p -values is handy for some of the more complicated algorithms of the next section.

Let $\mathbf{x}$ indicate all the data available for testing the family of hypotheses $H_{01}, H_{02}, \dots, H_{0N}$, and let $\text{FWER}_\alpha(\mathbf{x})$ be a FWER level- α test procedure based on $\mathbf{x}$. The general definition of adjusted p -value for case i is an analog of (3.3),

$$\tilde{p}_i(\mathbf{x}) = \inf_{\alpha} \{H_{0i} \text{ rejected by } \text{FWER}_\alpha(\mathbf{x})\}. \quad (3.16)$$

³ Formula (3.14) demonstrates “strong control”: FWER is bounded by α no matter what the pattern of true and false null hypotheses might be; “weak control” refers to methods that control FWER only if *all* the null hypotheses are true.

As an example, the *Šidák procedure* improves on the Bonferroni bound (3.13) by rejecting those hypotheses H_{0i} for which

$$p_i \leq 1 - (1 - \alpha)^{1/N}. \quad (3.17)$$

The corresponding adjusted p -value is

$$\tilde{p}_i = 1 - (1 - p_i)^N. \quad (3.18)$$

Exercise 3.2 (a) Verify (3.18). (b) Show that (3.17) improves on (3.13) in the sense of making it easier to reject every H_{0i} at any given level α . (c) Show that the Šidák procedure is FWER_α if the p -values $p_1, p_2, \dots, p_N$ are statistically independent.

Bonferroni's bound does not depend on independence, so Šidák's procedure cannot be considered a general improvement.⁴ *Holm's procedure* is an example of a more elaborate testing strategy that is in fact a general improvement: let the ordered p -values be denoted by

$$p_{(1)} \leq p_{(2)} \leq p_{(3)} \leq \dots \leq p_{(N)}, \quad (3.19)$$

and reject $H_{0(i)}$, the hypothesis corresponding to $p_{(i)}$, if

$$p_{(j)} \leq \frac{\alpha}{N - j + 1} \quad \text{for } j = 1, 2, \dots, i. \quad (3.20)$$

The next section shows that (3.19)–(3.20) is FWER_α . It is more powerful than Bonferroni's procedure since the rejection regions (3.20) are larger than the Bonferroni regions (3.13).

Exercise 3.3 Show that the adjusted p -value for Holm's procedure is

$$\tilde{p}_{(i)} = \max_{j \leq i} \left\{ (N - j + 1) p_{(j)} \right\}_1 \quad (3.21)$$

where $\{x\}_1 \equiv \min(x, 1)$ as before.

Adjusted p -values share the p -value virtue of not requiring a rigid pre-definition of the rejection level α . We can compute the $\tilde{p}_i$ values directly, as in (3.18) or (3.21), and see how “significant” each case is. The quotes are necessary here because $\tilde{p}_i$ usually does *not* follow a $\mathcal{U}(0, 1)$ distribution, or any other fixed distribution, when H_{0i} is true, so there is no universal interpretive scale such as Fisher's scale in Table 3.1.

⁴ The validity of Šidák's procedure can be extended beyond independence, including to multivariate normal testing situations.

Exercise 3.4 Suppose that all N hypotheses H_{0i} are true and that the N p -values are mutually independent. Calculate the distribution of $\tilde{p}_{(1)} = \{Np_{(1)}\}_1$. What is the limiting distribution as N goes to infinity?

3.3 Stepwise Algorithms

FWER control is a hard-line frequentist concept, far removed from the more relaxed empirical Bayes methods that are our main interest here. Nevertheless, it continues to play an important role in the literature of large-scale significance testing. Stepwise algorithms, developed mainly in the late 1980s, represent the most successful, and ingenious, attack on FWER control. This section and the next review some of the methodology, with no attempt at completeness and only hints of the theoretical ideas involved. We return to the empirical Bayes world in Chapter 4, where the results here will be viewed in contrast to false discovery rate control methods.

A *step-down* procedure begins with the ordered p -values $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(N)}$ as in (3.19), and from them defines a testing algorithm such that $H_{0(i)}$ can be rejected only if first $H_{0(j)}$ is rejected for $j = 1, 2, \dots, i-1$. In other words, if $p_{(i)}$ is small enough to cause rejection, then so must be $p_{(1)}, p_{(2)}, \dots, p_{(i-1)}$. Another way to say this is that the adjusted p -values are non-decreasing,

$$\tilde{p}_{(1)} \leq \tilde{p}_{(2)} \leq \dots \leq \tilde{p}_{(N)}, \quad (3.22)$$

so that $\tilde{p}_{(i)} \leq \alpha$ implies $\tilde{p}_{(j)} \leq \alpha$ for $j = 1, 2, \dots, i-1$. Step-down methods allow improvements on single-step procedures like (3.17). It can be shown, for example, that the step-down version of Šidák's procedure replaces (3.18) with

$$\tilde{p}_{(i)} = \max_{j \leq i} \left\{ 1 - (1 - p_{(j)})^{N-j+1} \right\}. \quad (3.23)$$

Holm's method (3.20)–(3.21) was one of the first examples of a step-down procedure. Here is the proof that it satisfies the FWER control property.

Proof Let I_0 be the set of indices corresponding to true null hypotheses H_{0i} , $N_0 = \#I_0$ the number of members of I_0 , and $i_0 = N - N_0 + 1$. Also let $\hat{i}$ be the stopping index for Holm's procedure, i.e., the maximum index

satisfying (3.20). The event

$$\begin{aligned} A = \left\{ p_{(i)} > \frac{\alpha}{N_0} \text{ for all } i \in I_0 \right\} &\implies \left\{ p_{(i_0)} > \frac{\alpha}{N_0} = \frac{\alpha}{N+1-i_0} \right\} \\ &\implies \left\{ \hat{i} < i_0 \right\} \implies \left\{ p_{(\hat{i})} < \frac{\alpha}{N_0} \right\} \equiv B. \end{aligned} \quad (3.24)$$

However, the Bonferroni bound shows that $\Pr\{A\} \geq 1 - \alpha$, and B implies that none of the true null hypotheses have been rejected. $\square$

Exercise 3.5 Explicate in detail the three steps in (3.24) and the conclusion that follows.

Holm's procedure illustrates a *closure principle* which is worth stating separately. Let I be a subset of the indices $\{1, 2, \dots, N\}$, and $\mathcal{I}$ the statement that *all* of the null hypotheses in I are true,

$$\mathcal{I} = \bigcap_I H_{0(i)}. \quad (3.25)$$

If I' is a larger subset, $I' \supseteq I$, then logically $\mathcal{I}' \Rightarrow \mathcal{I}$. Suppose that for every subset I we have a level- α non-randomized test function $\phi_I(\mathbf{x})$: $\phi_I(\mathbf{x})$ equals 1 or 0, with 1 indicating rejection of $\mathcal{I}$, satisfying

$$\Pr_{\mathcal{I}}\{\phi_I(\mathbf{x}) = 1\} \leq \alpha.$$

Now consider the simultaneous test function

$$\Phi_I(\mathbf{x}) = \min_{I' \supseteq I} \{\phi_{I'}(\mathbf{x})\}; \quad (3.26)$$

$\Phi_I(\mathbf{x})$ defines a rule that rejects $\mathcal{I}$ if and only if $\mathcal{I}'$ is rejected at level α for every I' containing I . But if $\mathcal{I}$ is true then $I \subseteq I_0$, the set of all true $H_{0(i)}$, and

$$\Pr_{I_0}\{\phi_{I_0}(\mathbf{x}) = 1\} \leq \alpha \quad \text{implies} \quad \Pr_{\mathcal{I}}\{\Phi_I(\mathbf{x}) = 1\} \leq \alpha. \quad (3.27)$$

In other words, the test Φ_I simultaneously controls the probability of rejecting any true subset I at level α .

The closure principle can be used to extend Bonferroni's bound to Holm's procedure. Let $I_i = \{i, i+1, \dots, N\}$. In terms of the ordered p -values (3.19), Bonferroni's rule rejects $\mathcal{I}_i$ at level α if $p_{(i)} \leq \alpha/(N+1-i)$. Note that $I_j \supseteq I_i$ for $j \leq i$.

Exercise 3.6 Complete the proof that Holm's procedure is FWER_α .

All of our calculations have so far begun with the simple Bonferroni bound (3.13). If we are willing to assume *independence* among the original p -values $p_1, p_2, \dots, p_N$, then a better bound, known as *Simes' inequality*, is available: when *all* the null hypotheses are true, then

$$\Pr \left\{ p_{(i)} \geq \frac{\alpha i}{N} \text{ for } i = 1, 2, \dots, N \right\} \geq 1 - \alpha, \quad (3.28)$$

with equality if the test statistics are continuous.

The proof of (3.28) begins by noting that in the independent continuous case, with all $H_{0(i)}$ true, $p_{(1)}, p_{(2)}, \dots, p_{(N)}$ are the order statistics of N independent $\mathcal{U}(0, 1)$ variates, as in (3.5). A standard order statistic result then shows that given $p_{(N)}$, the largest p -value, the ratios

$$(p_{(1)}/p_{(N)}, p_{(2)}/p_{(N)}, \dots, p_{(N-1)}/p_{(N)}) \quad (3.29)$$

are distributed as the order statistics from $(N - 1)$ independent $\mathcal{U}(0, 1)$ variates, while $p_{(N)}$ has cdf $p_{(N)}^N$.

Exercise 3.7 Use induction to verify Simes' inequality.

Starting from Simes' inequality, Hochberg used the closure principle to improve (i.e., raise) Holm's adjusted p -values (3.21) to

$$\tilde{p}_{(i)} = \min_{j \geq i} \left\{ (N - j + 1) p_{(j)} \right\}_1, \quad (3.30)$$

with $\{x\}_1 \equiv \min(x, 1)$. This is not a general improvement though, since Simes' inequality depends on independence of the test statistics. (Some progress has been made in broadening its validity.) Algorithms such as (3.30), whose definitions depend on the *upper* tail of the sequence $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(N)}$, are called "step-up procedures."

Exercise 3.8 Holm's step-down procedure (3.20) starts with $i = 1$ and keeps *rejecting* $H_{0(i)}$ until the first time $p_{(i)} > \alpha/(N - i + 1)$. Show that Hochberg's step-up procedure starts with $i = N$ and keeps *accepting* $H_{0(i)}$ until the first time $p_{(i)} \leq \alpha/(N - i + 1)$. This shows that Hochberg's procedure is more powerful than Holm's, i.e., rejects more often at the same α level. The step-up/step-down nomenclature is unfortunate here.

3.4 Permutation Algorithms

The prize property of the Bonferroni bound, that it holds true regardless of the dependence structure of the data, puts it at a disadvantage if we happen to know the structure. Westfall and Young proposed step-down procedures

that use permutation calculations to estimate dependence relationships, and then employ the estimated structure to improve on Holm's procedure.

Starting with the ordered p -values $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(N)}$, as in (3.19), let $r_1, r_2, \dots, r_N$ indicate the corresponding original indices,

$$p_{(j)} = p_{r_j}, \quad j = 1, 2, \dots, N. \quad (3.31)$$

Define

$$R_j = \{r_j, r_{j+1}, \dots, r_N\} \quad (3.32)$$

and

$$\pi(j) = \Pr_0 \left\{ \min_{k \in R_j} (P_k) \leq p_{(j)} \right\}. \quad (3.33)$$

Here $(P_1, P_2, \dots, P_N)$ indicates a hypothetical realization of the unordered p -values $(p_1, p_2, \dots, p_N)$ obtained under the *complete null hypothesis*⁵ $\mathbf{H}_0$ that *all* of the H_{0i} are true; (3.33) is computed with $p_{(j)}$ fixed at its observed value. The *Westfall–Young step-down min- p* adjusted p -values are then defined by

$$\tilde{p}_{(i)} = \max_{j \leq i} \{\pi(j)\}. \quad (3.34)$$

To see the connection with Holm's procedure, notice that Boole's inequality implies

$$\pi(j) \leq \sum_{k \in R_j} \Pr_0 \{P_k \leq p_{(j)}\} = (N - j + 1)p_{(j)}. \quad (3.35)$$

Comparing (3.34) with (3.21) shows that the Westfall–Young adjusted p -values are smaller than Holm's values. The proof that (3.34) satisfies the FWER control property, that is, that

$$\Pr \{\tilde{p}_i > \alpha \text{ for all } i \in I_0\} \geq 1 - \alpha, \quad (3.36)$$

for I_0 indexing the set of true hypotheses, is similar to the closure argument for Holm's procedure preceding Exercise 3.6. It does, however, involve an extra assumption, called “subset pivotality”: that the vector $(P_i, i \in I_0)$ always follows the distribution it has under the complete null $\mathbf{H}_0$. In other words, the fact that some of the cases are non-null does not affect the joint distribution of the null p -values.

The min- p procedure can be difficult to implement. Westfall and Young also proposed a simpler variant, called “max- T .” Let

$$t_{(1)} \geq t_{(2)} \geq \dots \geq t_{(N)} \quad (3.37)$$

⁵ $\Pr_0$ now indicates probabilities computed under $\mathbf{H}_0$.

indicate ordered values of the original test statistics that gave the p -values (the two-sample t -statistics (2.2) for the prostate and DTI studies), with ordered values $t_{(j)} = t_{r_j}$. Also let $(T_1, T_2, \dots, T_N)$ represent a hypothetical unordered realization obtained under the complete null hypothesis H_0 . Now define⁶

$$\pi(j) = \Pr \left\{ \max_{k \in R_j} (T_k) \geq t_{(j)} \right\}, \quad (3.38)$$

yielding adjusted p -values $\tilde{p}_{(i)}$ as in (3.34). If all T_i have the same cdf $F(T)$ then $p_i = 1 - F(t_i)$ and (3.38) is the same as (3.33), but otherwise the two procedures differ.

How can we evaluate $\pi(j)$ in (3.33) or (3.38)? In some situations, permutation methods provide straightforward and plausible answers. Consider the prostate data (2.1): the data matrix X is 6033×102 , with its first 50 columns representing the healthy controls and the last 52 columns the cancer patients. X gave the 6033-vector $\mathbf{t}$ of two-sample t -statistics via row-wise application of (2.2)–(2.3).

Now let X^* be a version of X in which the columns have been randomly permuted: formally, if $\mathbf{J}^* = (j_1^*, j_2^*, \dots, j_n^*)$ is a randomly selected permutation of $(1, 2, \dots, n)$ then X^* has entries

$$x_{ij}^* = x_{iJ^*(j)} \quad \text{for } j = 1, 2, \dots, n \text{ and } i = 1, 2, \dots, N. \quad (3.39)$$

Applying calculations (2.2)–(2.3) to X^* (and still considering the first 50 columns as controls and the last 52 as cancer patients) yields a 6033-vector of “permutation t -values”

$$\mathbf{T}^* = (T_1^*, T_2^*, \dots, T_N^*)'. \quad (3.40)$$

Independently repeating the permutation process some large number B times allows us to estimate (3.38) by simply counting the proportion of times $\max(T_k^*, k \in R_j)$ exceeds $t_{(j)}$,

$$\hat{\pi}(j) = \# \left\{ \max_{k \in R_j} (T_k^*) > t_{(j)} \right\} / B. \quad (3.41)$$

Here R_j and $t_{(j)}$ retain their original values, only the T_k^* vary. Finally, the adjusted p -values $\tilde{p}_{(i)}$ are estimated from the $\hat{\pi}(j)$ as in (3.34). (The min- p calculation is more difficult to implement, explaining the greater popularity of max- T .)

The key idea is that the permutation distribution of $\mathbf{T}^*$ is a reasonable

⁶ We could just as well define $\pi(j)$ with respect to two-sided or left-sided versions of (3.37)–(3.38).

stand-in for the hypothetical distribution of T we would obtain under the complete null hypothesis H_0 . This is plausible since

- permuting the columns of X destroys any true differences between controls and cancer patients, thereby enforcing H_0 ;
- and since columns are permuted intact, the correlation structure between the rows (i.e., the genes) is maintained.

Permutation methods have played a major role in the large-scale testing literature. We will be discussing their virtues and limitations in several upcoming chapters.

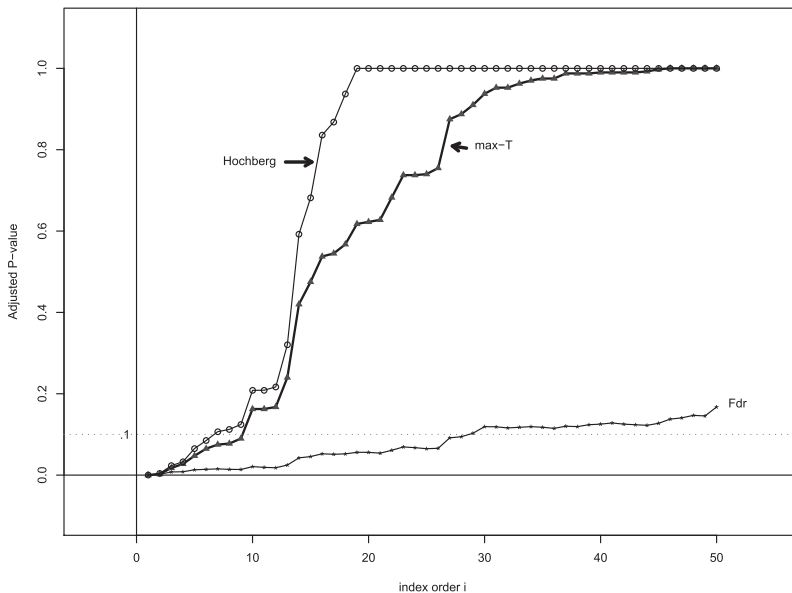


Figure 3.2 Adjusted p -values $\tilde{p}_{(i)}$ for the prostate data (right-sided), $N = 6033$; circles are Hochberg step-up procedure (3.30), triangles are max- T procedure (3.34)–(3.38). Also shown are estimated false discovery rates (3.42).

Figure 3.2 shows the max- T adjusted p -values $\tilde{p}_{(i)}$ for the prostate data. Here they are seen to be a substantial improvement over the Hochberg step-up values (3.30) (which are the same as Holm's values (3.21) in this case). For instance, the first nine of the max- T $\tilde{p}_{(i)}$ values are less than 0.1, versus only six for Hochberg.

Also shown in Figure 3.2 are the estimated false discovery rates $\overline{\text{Fdr}}_{(i)}$,

$$\overline{\text{Fdr}}_{(i)} = N \cdot [1 - \Phi(z_{(i)})] / \#\{z_j \geq z_{(i)}\} \quad (3.42)$$

where $z_{(1)} \geq z_{(2)} \geq \dots \geq z_{(N)}$ are the ordered z -values and Φ the standard normal cdf. These follow definitions (2.23)–(2.24) with $\mathcal{Z} = [z_{(i)}, \infty)$, and π_0 taken to equal its upper bound 1. The striking fact is how much smaller $\overline{\text{Fdr}}_{(i)}$ is than either version of $\tilde{p}_{(i)}$. For $i = 20$, $\overline{\text{Fdr}}_{(20)} = 0.056$, while $\tilde{p}_{(20)} = 0.62$ for $\max\text{-}T$. These results are not contradictory: there is a good chance that at least one of the 20 genes is null, but the expected number of nulls (0.056×20) is not much bigger than 1. The Fdr criterion has become popular because its more liberal conclusions are a good fit to modern applications having N in the thousands.

With $N = 6033$ and $i = 20$, as in Figure 3.2, Holm's method (3.20) and Hochberg's procedure are almost the same as the Bonferroni bound (3.13). Of course there are still plenty of small- N multiple testing problems, where these more sophisticated procedures come into their own.

3.5 Other Control Criteria

FWER control dominates the traditional multiple comparison literature, but other criteria have been proposed. Two other examples of quantities to control are the *per comparison error rate*

$$\text{PCER} = E\{\text{Number true null hypotheses rejected}\}/N \quad (3.43)$$

and the *expected error rate*

$$\text{EER} = E\{\text{Number wrong decisions}\}/N, \quad (3.44)$$

a wrong decision being rejection of H_{0i} when it should be accepted or vice versa. Neither PCER nor EER have attracted the attention accorded FWER.

The microarray era, with case sizes zooming up from $N = 10$ to $N = 10\,000$, has brought dissatisfaction with FWER control. Less conservative methods that still command scientific respectability are being developed, an interesting example being Lehmann and Romano's k -FWER criteria, which aims to control the probability of rejecting k or more true null hypotheses; $k = 1$ is the usual FWER, but choosing a larger value of k gives more generous results.

A simple extension of the Bonferroni bound (3.13) provides k -FWER control.

Theorem 3.1 The procedure that rejects only those null hypotheses H_{0i} for which

$$p_i \leq k\alpha/N \quad (3.45)$$

controls k -FWER at level α ,

$$\Pr\{k \text{ or more true } H_{0i} \text{ rejected}\} \leq \alpha. \quad (3.46)$$

Proof Let I_0 index the true null hypotheses H_{0i} as in (3.14), with $N_0 = \#I_0$, and let $N_0(\text{rej})$ be the number of falsely rejected H_{0i} . Then

$$\begin{aligned} \Pr\{N_0(\text{reg}) \geq k\} &\leq E\{N_0(\text{reg})\}/k = \sum_{i \in I_0} \Pr\{p_i \leq k\alpha/N\}/k \\ &= \sum_{i \in I_0} \frac{k\alpha/N}{k} \\ &= \frac{N_0}{N} \alpha \leq \alpha. \quad \square \end{aligned} \quad (3.47)$$

Exercise 3.9 Verify the first inequality above (*Markov's inequality*).

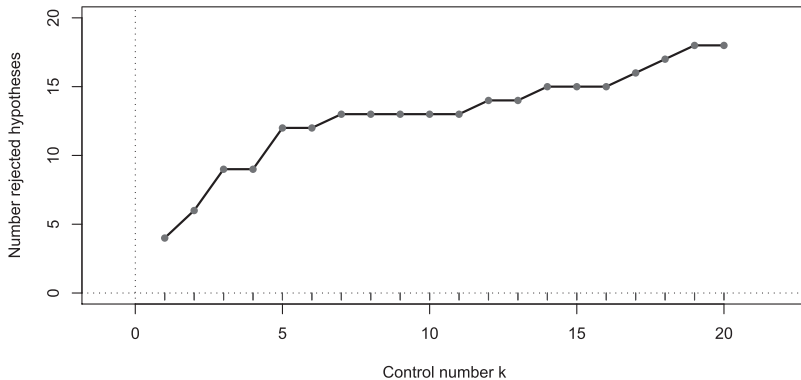


Figure 3.3 Number of rejections by k -FWER control criterion (3.45), $\alpha = 0.05$, for $k = 1, 2, \dots, 20$; prostate data (right side).

Figure 3.3 applies k -FWER control, $\alpha = 0.05$, to the right side (that is, the positive z -values) of the prostate data. The graph traces the number of genes satisfying rejection criterion (3.45) for different choices of k , ranging from $k = 1$ to $k = 20$. At $k = 1$, the usual FWER criterion, four genes are rejected, climbing to 18 rejections at $k = 20$.

Once again the comparison with estimated false discovery rates (3.42) is startling: $\widehat{\text{Fdr}}_{(18)} = 0.052$, so the estimated number of true null hypotheses among the first 18 rejected is less than one, implying k -FWER hasn't taken much advantage of its allowance of $k = 20$ errors; probably it has made only one or two. One of the main goals of Chapter 4 is to understand why FDR control permits such liberal conclusions.

Notes

Westfall and Young's (1993) book was a notable successor to Miller (1981), showing how modern computation could be used to good effect on multiple testing problems. Dudoit and van der Laan's (2008) book takes another large step in the direction of computation-intensive testing algorithms.

A series of ingenious papers produced the step-down and step-up algorithms of Section 3.3 and Section 3.4: Holm (1979), Simes (1986), Hommel (1988), and Hochberg (1988). The closure principle is nicely stated in Marcus et al. (1976), though its origins go back further. Dudoit et al. (2003) provide an excellent review of the whole theory.

An influential example of applying ANOVA methods to microarray data matrices appears in Kerr et al. (2000). Storey's *optimal discovery procedure* (2007) can be considered as an implementation of EER control (3.44). Efron and Gous (2001) provide a discussion of Fisher's interpretive scale for p -values, Table 3.1, and its Bayes-factor competitor, Jeffrey's scale. Lehmann and Romano (2005a) and Romano et al. (2008) discuss other control criteria besides k -FWER.